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An extended Tennessee Eastman simulation dataset for fault-detection and decision support systems

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ABSTRACT

The Tennessee Eastman Process (TEP) is a frequently used benchmark in chemical engineering research. An extended simulator, published in 2015, enables a more in-depth investigation of TEP, featuring additional, scalable process disturbances as well as an extended list of variables. Even though the simulator has been used multiple times since its release, the lack of a standardized reference dataset impedes direct comparisons of methods. In this contribution we present an extensive reference dataset, incorporating repeat simulations of healthy and faulty process data, additional measurements and multiple magnitudes for all process disturbances. All six production modes of TEP as well as mode transitions and operating points in a region around the modes are simulated. We further perform fault-detection based on principal component analysis combined with T^2 and Q charts using average run length as a performance metric to provide an initial benchmark for statistical process monitoring schemes for the presented data.

1. Introduction

The recent decades have seen a major push towards the optimization of process operations. Along with this go multiple areas of research within the process monitoring and fault-detection and isolation (FDI) communities. The existing approaches can broadly be classified into model-based, data-driven and hybrid approaches. Model-based approaches based on first-principle models require a high degree of a priori knowledge about the system under investigation. Their application to large-scale systems can thus often be challenging, especially if parts of these systems are subject to uncertainty. Data-driven approaches use available process measurements to classify the operation of a system. Typically, intervals during which the operation is known to be normal are used as training data to characterize the nominal operation of the process so that faults can be detected as deviations from the learned nominal process characteristics. Fault isolation using data-driven approaches often requires additional data representing the characteristic behaviour of the process while it is subjected to faults. To compare performances of different FDI approaches, realistic benchmark cases are required.

The Tennessee Eastman process (TEP) has been extensively used for this purpose since the initial publication by Downs and Vogel (1993). It provides a benchmark case that can be used to investigate most challenges continuous processes present. In the original publication, Downs and Vogel (1993) list plant-wide control design, multivariate control, optimization, predictive control, adaptive control, nonlinear control, process diagnostics and educational purposes as potential applications of the simulator. Different control-designs (Ricker and Lee, 1995; Ricker, 1996; Larsson et al., 2001; Lyman and Georgakis, 1995) were proposed in the years following the initial publication of the process and others have been added at a later stage (Molina et al., 2011). Many recent publications involving the Tennessee Eastman process focus on fault detection using classical statistics or machine learning based approaches. Arunthavanathan et al. (2020) presents an unsupervised learning method for the detection and diagnosis of unknown fault-conditions by integrating a unary classification algorithm for fault-detection with a dynamic shallow neural network for fault-classification. Han et al. (2020) iteratively optimize the number of hidden layers in a Long short-term memory (LSTM) network to achieve higher diagnostic accuracy. Maran Beena and Pani (2020) apply Gaussian process regression (GRP)-based process monitoring to the Tennessee Eastman process and investigate the effect of GPR hyper parameters on monitoring efficiency, while Fezai et al. (2020) combine GPR with a generalized likelihood ratio test. The performance of Kantorovich Distance Independent Com-

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ponent Analysis is investigated by Kini and Madakyaru (2020), who also provide a comparison to other frequently used statistical monitoring techniques. Dong and Qin (2018) present a fault-diagnosis method based on deep convolutional neural networks to detect 20 process faults. Ragab et al. (2018) apply a machine-learning based classification technique called logical analysis of data to extract interpretable patterns linked to physical phenomena to the base case of TEP. Bakdi and Kouadri (2018) presents a process monitoring scheme that is based on the integration of univariate and multivariate statistical analysis methods by using an adaptive threshold scheme instead of conventional fixed control limits. Zhang and Zhao (2017) present a deep belief network for the detection of process faults in TEP and compare the performance of one-class-one-network (OCON) and one-variable-one-network (OVON) architectures regarding training complexity and classification accuracy. Recent years have also seen a number of publications using the Tennessee Eastman process within the field of operations research and the design of decision support systems. These publications focus on methods that yield interpretable results for process diagnostics and operation, capable of explaining the recommendations and decisions of AI-based systems. Venkatasubramanian (2019) stresses the need for such systems for fault-diagnosis and hazard analysis. Suresh et al. (2019) present a semi-automated method for the generation of hierarchical causal models of continuous processes using transfer entropy as a causality measure, while Mori et al. (2014) use a Bayesian approach to generate a causal network from process data. Gao et al. (2016) present an approach to extract meaningful patterns highlighting the relations of variables across streams and units using sparse principal component analysis. He et al. (2014) present a hybrid fault-diagnosis method that integrates reconstruction-based multivariate contribution analysis with fuzzy signed directed graphs for root-cause analysis. Further research opportunities, especially with regard to the possibilities opened up by a new simulator by Bathelt et al. (2015), are outlined by Capaci et al. (2018).

The amount of publications featuring TEP in recent years is a good indicator for its continued relevance in various research communities. A hindrance to the comparison of the contributions published since 2015 is that the presented methods and performance analysis are conducted based on different process data for TEP, however. Some contributions use the simulator by Bathelt et al. (2015) to generate process data while others utilize existing datasets, which are generated using older simulators and which feature different control structures for TEP. In order to facilitate the use of data from the simulator by Bathelt et al. (2015), it is necessary to provide a reference dataset generated using said simulator, which makes use of the newly introduced functionalities and provides a proper benchmark for future research, just as currently available datasets provide a reference benchmark for previous iterations of TEP simulators.

1.1. Existing simulators

The original model, consisting of a set of Fortran routines, was presented by Downs and Vogel (1993) and can be found on various sources online. It describes the open-loop behaviour of the process, which is unstable. The repository by Braatz (2020) includes a simulator based on a set of Fortran routines that implements the control scheme presented by Lyman and Georgakis (1995), which allows the simulation of 20 process faults at full fault-magnitude for the base case. Sample and fault-activation times can be adjusted as required and seeds for the random generator can be set explicitly. While most of the datasets currently used for fault detection and diagnosis are based on this simulator, it has not been extended as other simulators mentioned below and by now offers fewer options for modifications of the simulation. Gins et al. (2014) integrate the Tennessee Eastman model into the RAYMOND simulation package, thus extending simulation capabilities to facilitate the inclusion of input variations, different noise signatures and additional process faults. Other implementations of control-schemes that stabilize the process are available at the Tennessee Eastman Challenge Archive (Ricker, 2020). These include a model-predictive control (MPC) approach by Ricker and Lee (1995) and a decentralized control approach from Ricker (1996). Additionally, a MPC approach for a simplified model of the Tennessee Eastman process from Ricker (1993) and a summary of optimal steady states of the TEP for the six operating modes, presented in Ricker (1995), are included. Most notably, the Challenge Archive includes a revised model of the TEP by Bathelt et al. (2015). The revised model includes implementations of three control strategies. The models labeled *Multiloop_mode1* and *Multiloop_mode3* implement the decentralized control strategy by Ricker (1996), without and with override loops, respectively, while *Multiloop_Skoge* implements the control strategy presented by Larsson et al. (2001). The revised model introduces multiple significant improvements compared to previous versions (Bathelt et al., 2015). Most importantly, it remedies an issue where the simulation results show a dependency on the chosen solver and simulation time-increment in cases where disturbances of the type *random variation* are activated. Further, eight new process disturbances (IDV 21–28 in Table 3) of the type *random variation* as well as the option to scale the magnitude of disturbances are added. The revised model additionally includes an extended set of process measurements as well as the option to log internal process variables.

Table 1
Steady-state process operation modes of the Tennessee Eastman Process Downs and Vogel (1993).

Mode	G/H mass ratio	Production (Stream 11)
1	50/50	14076 kg/h
2	10/90	14076 kg/h
3	90/10	11111 kg/h
4	50/50	maximum
5	10/90	maximum
6	90/10	maximum

1.2. Existing datasets

While the investigation of the efficacy of different control schemes requires direct use of a TEP simulator, most process analysis and FDI research only requires a sufficiently complete database of simulation results for a given control scheme. The obvious benefits of a database are direct data access and the ability to compare different methods using identical baselines, meaning identical data. Braatz (2020) provides training and test data for 21 simulated faults, which are summarized in Chiang et al. (2000). Each fault and the normal process operation in mode 1 of the process are simulated with a sampling rate of 3 minutes. Training data files contain runs of 25 hours and testing data files runs of 48 hours, resulting in 500 and 960 observations for each run, respectively. A similar data set, though updated more recently, is provided by Chen (2019). The main drawback of the data provided by Braatz (2020) is that each fault case is represented by a single simulation, providing a single noise signature for each given run and not allowing evaluations of performance metrics such as the average run-length that require multiple simulations of the same event. This issue is addressed by Rieth et al., 2017, who simulate healthy data and the process faults with the same simulation setup as Braatz (2020) (sampling rate of 3 minutes for a duration of 25 hours for training and 48 hours for test data, respectively), but run each simulation

Table 2
Setpoints for the control-scheme by Ricker (1996) for optimal steady-state operation as defined by Ricker (1995).

Setpoint label	Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	Mode 6
Production	22.89	22.73	18.04	36.04	23.55	20.2
Stripper level	50	50	50	50	50	50
Seperator level	50	50	50	50	50	50
Reactor level	65	65	65	65	65	65
Reactor pressure	2800	2800	2800	2800	2800	2800
Mol % G	53.8	11.66	90.09	53.35	11.65	90.07
yA	63.137	64.196	62.11	61.94	64.03	61.468
yAC	51	54.24	47.43	58.76	54.32	48.79
Reactor temperature	122.9	124.2	121.9	128.2	124.6	123
Recycle valve pos.	1	1	77.62	1	1	71.166
Steam valve pos.	1	1	1	1	1	1
Agitator setting	100	100	100	100	100	100

500 times with independent, non-overlapping seeds for the random number generator, allowing a thorough analysis of the different process faults in operating mode 1. Manca (2020) provides reference data for alarm management studies of TEP in Mode 1 including simulation data and corresponding alarm data for the seven pre-defined process faults of TEP that can be described as step changes. Using the simulator by Bathelt et al. (2015), these faults are simulated at different fault-magnitudes. For investigation of faults at full magnitude in production mode 1, Rieth et al. (2017) currently provides the most complete dataset, viewed from a statistical process control perspective, because it provides multiple simulation runs for identical simulation settings with different seeds for the random number generator for 20 process faults.

1.3. Novelty and contribution

We present a new reference dataset for the Tennessee Eastman Process that extends existing reference data in both scope, regarding its potential applications, and scale, in terms of the amount of generated data. Contrary to previously presented data, which focused exclusively on a specific operating mode of the process (Mode 1), we present simulations for all 28 process faults that are part of the latest TEP simulator by Bathelt et al. (2015) for all six operating modes that were originally presented by Downs and Vogel (1993). We further increase the simulation run-time for all simulations to 100 hours, so that it is guaranteed that both transient as well as stationary effects of the simulated faults are captured in each run. Additionally, we provide a benchmark for multivariate fault detection performance for the presented dataset using average run length of PCA-based Hotelling T^2 and Q charts as the performance metric for all modes and simulated faults.

In addition to simulations for the purpose of developing and testing the performance of fault detection and isolation schemes, we present data that can be used to test the robustness of such schemes for processes with varying nominal operating conditions. Further, the effects of deliberate set-points changes for all six operating modes as well as transitions between the specified operating modes using different transients are simulated. Making use of the extended simulation capabilities of the simulator presented by Bathelt et al. (2015), the logging data is extended to include 53 additional data points for each time step, representing additional process measurements and internal process variables.

2. Tennessee eastman process

The process involves the production of two liquid product components G and H from four gaseous reactants A, C, D and E with an additional inert B and a byproduct F. The reaction system consists of four exothermic and irreversible reactions which are described by:

$$A(g) + C(g) + D(g) \rightarrow G(liq) \qquad \text{(Product 1)}$$
$$A(g) + C(g) + E(g) \rightarrow H(liq) \qquad \text{(Product 2)}$$
$$A(g) + E(g) \rightarrow F(liq) \qquad \text{(Byproduct)}$$
$$3D(g) \rightarrow F(liq) \qquad \text{(Byproduct)}$$

The five major process units are the reactor, product condenser, vapor-liquid separator, recycle compressor and product stripper. The reactants A, D and E flow to the reactor from upstream. The reactor output stream, consisting of products and unreacted feed, is fed into a condenser and subsequent vapor-liquid unit. Condensed components enter the product stripper where the remaining reactants are separated from the products G and H and non-condensed components are recycled to the reactor feed through the recycle compressor. Fig. 1 shows a diagram of the process and the available process measurements. Six different process operation modes which are distinguished by the G/H mass ratio and the production rate are introduced by Downs and Vogel (1993). The definition of the modes in regard to desired mass-ratios and production rates are summarized in Table 1. Ideal steady-state operating conditions for all modes are investigated by Ricker (1995) and the respective control-setpoints for the control strategy by Ricker (1996) are listed in Table 2. A complete list of the measured and manipulated variables for the six modes can be found in Ricker (1995).

3. Dataset for FDI and decision support

The presented dataset is a reference dataset for studies within the fields of statistical process control, fault-detection and identification and operations research and is freely accessible at Reinartz et al. (2021). It was generated using a modified version of the simulator provided by Bathelt et al. (2015). The simulations using the *Multiloop_mode3* simulink model enable simulation in all six operating modes. The control setpoints are generated as step- or ramp-signals rather than constants, as implemented by Bathelt et al. (2015) to enable the simulation of setpoint- and mode-changes. The model flag activation is set to 227, resulting in the output of additional process measurements and internal variables related to IDVs of the type *random disturbance*, as well as the addition of white Gaussian measurement noise, as specified by Bathelt et al. (2015).

3.1. Simulation groups

The provided data can be separated into three different groups. The majority of the simulations belong to the simulation group for statistical process monitoring and other fault-detection and identification approaches. Simulations in this group consist of a run-

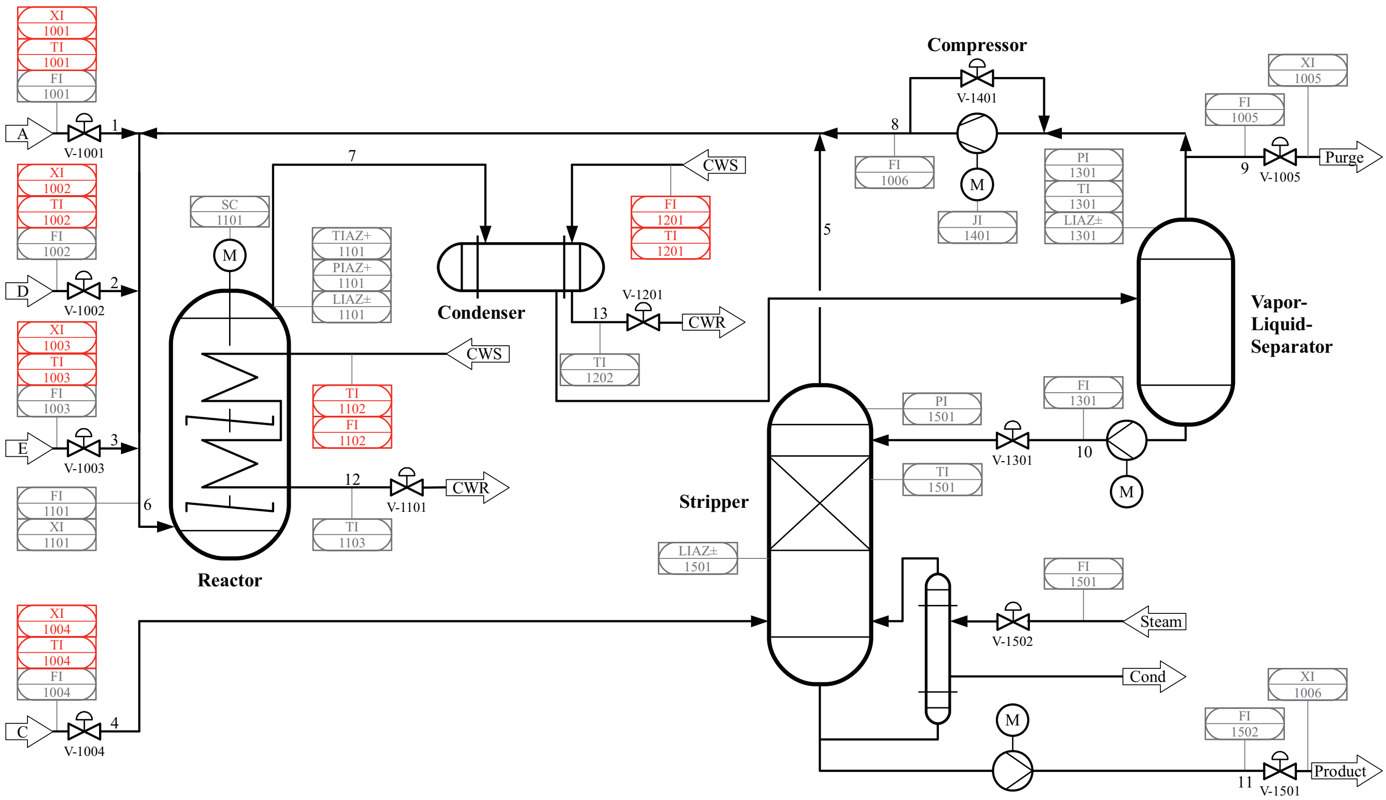

Fig. 1. P&ID of the revised Tennessee Eastman process model from Bathelt et al. (2015). Measurements introduced by Downs and Vogel (1993) in grey, additional measurements introduced by Bathelt et al. (2015) in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 3
Process faults for the revised Tennessee Eastman simulator.

Variable	Process Variable	Type
IDV(1)	A/C feed ratio, B composition constant (stream 4)	Step
IDV(2)	B composition, A/C ratio constant (stream 4)	Step
IDV(3)	D feed temperature (stream 2)	Step
IDV(4)	Water inlet temperature for reactor cooling	Step
IDV(5)	Water inlet temperature for condenser cooling	Step
IDV(6)	A feed loss (stream 1)	Step
IDV(7)	C header pressure loss (stream 4)	Step
IDV(8)	A/B/C composition of stream 4	Random variation
IDV(9)	D feed (stream 2) temperature	Random variation
IDV(10)	C feed (stream 4) temperature	Random variation
IDV(11)	Cooling water inlet temperature of reactor	Random variation
IDV(12)	Cooling water inlet temperature of separator	Random variation
IDV(13)	Reaction kinetics	Random variation
IDV(14)	Cooling water outlet valve of reactor	Sticking
IDV(15)	Cooling water outlet valve of separator	Sticking
IDV(16)	Variation coefficient of the steam supply of the heat exchange of the stripper	Random variation
IDV(17)	Variation coefficient of heat transfer in reactor	Random variation
IDV(18)	Variation coefficient of heat transfer in condenser	Random variation
IDV(19)	Unknown	Unknown
IDV(20)	Unknown	Random variation
IDV(21)	A feed (stream 1) temperature	Random variation
IDV(22)	E feed (stream 3) temperature	Random variation
IDV(23)	A feed flow (stream 1)	Random variation
IDV(24)	D feed flow (stream 2)	Random variation
IDV(25)	E feed flow (stream 3)	Random variation
IDV(26)	A and C feed flow (stream 4)	Random variation
IDV(27)	Reactor cooling water flow	Random variation
IDV(28)	Condenser cooling water flow	Random variation

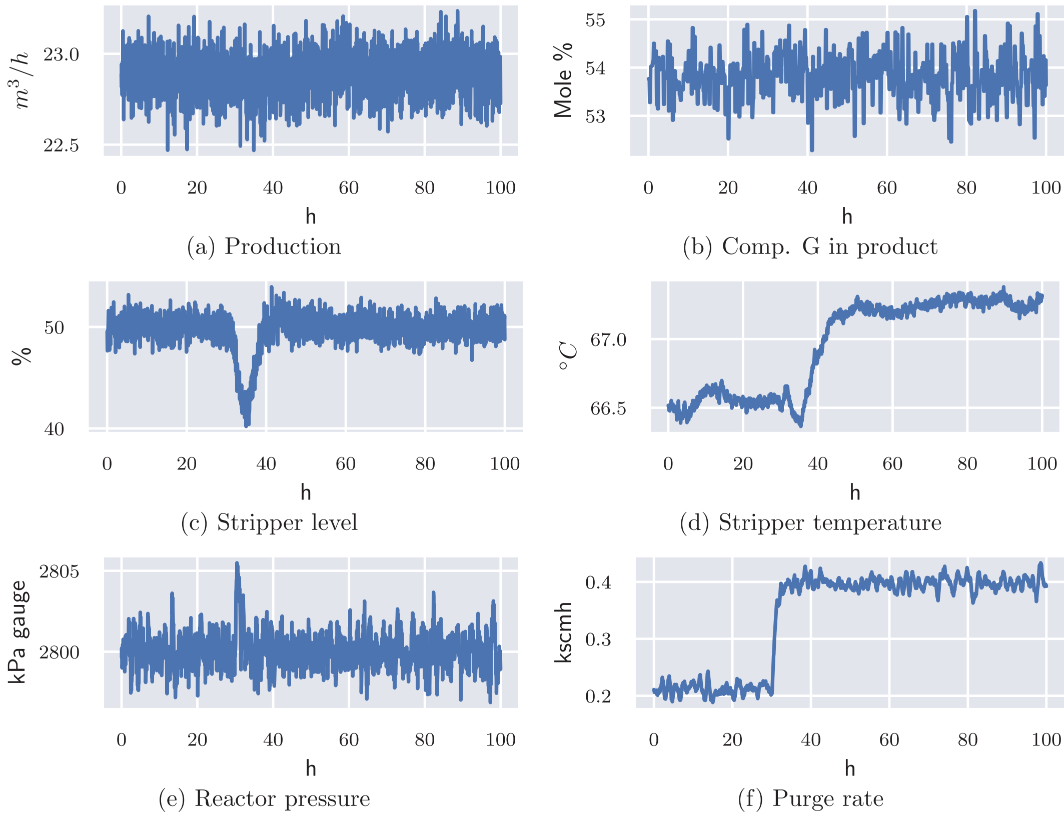

Fig. 2. Process response to activation of IDV2 at 100% fault magnitude after 30 simulation hours in operating mode 1.

in, during which the process operates nominally, followed by the introduction of a single fault into the system, which results in a reaction of the implemented control system and changed operating conditions and possibly an emergency shutdown of the plant. An example of a simulation from this simulation group is shown in Fig. 2, where the reaction of the process to the activation of the fault IDV2, described in Table 3, are presented. Considering the six operating modes in combination with the 28 process faults, the single fault simulations feature 168 different setups that are investigated. In addition to the investigation of the process faults at full magnitudes, we further simulate all faults at magnitudes 25%, 50% and 75%, leading to 672 different simulation setups. In order to create a dataset that is viable for the development and testing of statistical process monitoring approaches, each setup is simulated 200 times at 100% fault magnitude and 100 times at 25%, 50% and 75% fault magnitude, resulting in 500 simulations of each process fault and 84,000 fault simulations overall.

The second group contains simulations that feature deliberate set-point changes introduced to the six nominal operating modes. These set-point changes mimic the effect of actions performed by human operators by considering the 12 direct external access points, listed in Table 2, to the Tennessee Eastman process for the used control scheme. Following a run-in as in the first simulation group, single set-point variations in a 15% interval around the original value are introduced to the system. Given the chosen increment of set-point variations of 5% of the original value, this results in 7 simulation runs for each set-point and thus 504 simulation runs considering the six modes, 12 set-points and 20 magnitudes. Each set-point change is further simulated with different transition profiles; a step-profile and ramp profiles with ramping durations of 10, 20, 30 and 40 hours. We see two main uses of this simulation group. In the context of statistical process monitoring, it allows the investigation of the robustness to slightly changing, though nominal, operating conditions that should not be classified as faulty operations. This addresses the fact, that many currently presented approaches perform very well, as long as the nominal operating conditions don't vary too much. Especially for the Tennessee Eastman Process, most presented approaches are trained on a clearly defined, single operating point (Mode 1), and every process configuration that deviates is classified as a fault. Because process operating conditions can not be specified that precisely in many real applications, we believe that it is necessary to provide data for nominal operations that features more variation, including multiple feasible operating points as well as dynamic transitions featuring ramp and step profiles between such stationary operating conditions. In addition to its application to statistical process monitoring, the second group of simulations can be used for the investigation of the means of externally influencing the Tennessee Eastman process via the control set-points in the presence of closed loop control, mimicking a human control room operators means of influencing the process. Such investigations can be useful for research concerned with decision support systems for operators, as active decision support tools could propose the set-point changes in the provided simulation group as feasible measures in for example reconfiguration scenarios after activation of process faults.

The third simulation group contains simulations of transitions between the six operating modes, with ramp transition profiles for the respective set-points with ramping durations of 10, 20, 30 and 40 simulation hours. This group completes the data for nominal process operation, as any real process with six specified modes would also have to feature transitions between said modes.

3.2. Simulation parameters

Each simulation run is simulated for a duration of 100 simulation hours at a sampling rate of 3 minutes, resulting in 2000 obser-

Table 4
Variables contained in the 'processdata' dataset.

Legend	Type	Description	Legend	Type	Description
pd(1)	meas	Time (h)	pd(28)	meas	Component E to Reactor (mol %)
pd(2)	meas	A Feed (kscmh)	pd(29)	meas	Component F to Reactor (mol %)
pd(3)	meas	D Feed (kg/h)	pd(30)	meas	Component A in Purge (mol %)
pd(4)	meas	E Feed (kg/h)	pd(31)	meas	Component B in Purge (mol %)
pd(5)	meas	A and C Feed (kscmh)	pd(32)	meas	Component C in Purge (mol %)
pd(6)	meas	Recycle Flow (kscmh)	pd(33)	meas	Component D in Purge (mol %)
pd(7)	meas	Reactor Feed Rate (kscmh)	pd(34)	meas	Component E in Purge (mol %)
pd(8)	meas	Reactor Pressure (kPa gauge)	pd(35)	meas	Component F in Purge (mol %)
pd(9)	meas	Reactor Level (%)	pd(36)	meas	Component G in Purge (mol %)
pd(10)	meas	Reactor Temperature (° C)	pd(37)	meas	Component H in Purge (mol %)
pd(11)	meas	Purge Rate (kscmh)	pd(38)	meas	Component D in Product (mol %)
pd(12)	meas	Product Sep Temp (° C)	pd(39)	meas	Component E in Product (mol %)
pd(13)	meas	Product Sep Level (%)	pd(40)	meas	Component F in Product (mol %)
pd(14)	meas	Product Sep Pressure (kPa gauge)	pd(41)	meas	Component G in Product (mol %)
pd(15)	meas	Product Sep Underflow (m^3/h)	pd(42)	meas	Component H in Product (mol %)
pd(16)	meas	Stripper Level (%)	pd(43)	manip	D feed (%)
pd(17)	meas	Stripper Pressure (kPa gauge)	pd(44)	manip	E Feed (%)
pd(18)	meas	Stripper Underflow (m^3/h)	pd(45)	manip	A Feed (%)
pd(19)	meas	Stripper Temp (° C)	pd(46)	manip	A and C Feed (%)
pd(20)	meas	Stripper Steam Flow (kg/h)	pd(47)	manip	Compressor recycle valve (%)
pd(21)	meas	Compressor Work (kW)	pd(48)	manip	Purge valve (%)
pd(22)	meas	Reactor Coolant Temp (° C)	pd(49)	manip	Separator liquid flow(%)
pd(23)	meas	Separator Coolant Temp (° C)	pd(50)	manip	Stripper liquid flow (%)
pd(24)	meas	Component A to Reactor (mol %)	pd(51)	manip	Stripper steam valve (%)
pd(25)	meas	Component B to Reactor (mol %)	pd(52)	manip	Reactor Coolant (%)
pd(26)	meas	Component C to Reactor (mol %)	pd(53)	manip	Condenser Coolant (%)
pd(27)	meas	Component D to Reactor (mol %)	pd(54)	manip	Agitator speed (%)

vations. During the first 30 hours, the process is run in a nominal operating mode, meaning one of the six modes presented above. After 30 hours the process is subjected to either a process fault from the list presented in Table 3 or a control-setpoint change. The final 70 simulation hours represent the reaction of TEP to the introduced change. Downs and Vogel (1993) state that the process can show dynamic behaviour after a control-setpoint change for 24–48 hours before settling on a new stationary operating condition. Inspection of the dynamic behaviour of the process' reaction to setpoint changes shows that 48 hours is a very conservative measure and that a new stationary operating point is reached within 40 hours after the introduction of a deviation. Similarly, a transition time of 40 simulation hours is sufficient to capture the transient behaviour resulting from the presented process disturbances of the type 'Step'. Disturbances of the type 'Random Variation' and 'Sticking' affect the process behaviour continuously so that the process reaction can't be clearly separated into transient and stationary responses. Based on the analysis of maximum process response times, the intent behind the 70 simulation hours of fault simulation is that this interval can be split into 40 hours of simulation time which include the dynamic response and 30 hours of simulation time which represent stationary process behaviour under changed operating conditions. This 30-40-30 split of the 100 simulation hours is applied to both fault simulations and setpoint variation runs.

3.3. Data content

Three different datasets are provided for each simulation run. The first dataset, labeled "processdata", contains information about simulation time, the original set of process measurements and the manipulated variables. This data, summarized in Table 4, includes all variables that have been frequently used in statistical process control studies. The second dataset, labeled "additional_meas" contains the additional process measurements shown in Table 5. These values correspond to the first model flag introduced in the simulator by Bathelt et al. (2015). In the third dataset, labeled "idv_monitoring", specific information about selected random disturbances is provided. This data, summarized in Table 6, corresponds to the second model flag by Bathelt et al. (2015). It should be noted that the "idv_monitoring" data is only provided for simulations in which process faults of the type random variation (IDV 8–13, 16–18, 20–28) are active. It was decided to include the additional measurement points provided by Bathelt et al. (2015) in the data, because they provide an opportunity to study how a higher degree of instrumentation affects fault detection and identification in the process. The monitoring of the random-variation disturbances, for which the logged values are not affected by noise, makes it possible to study the effect of the disturbances in greater detail.

3.4. Storage format

To make the data access platform and application independent, the hdf5 data format is chosen for data storage, which provides APIs for most if not all programming languages frequently used for data analysis, such as R, Python, MATLAB® etc. The data is structured hierarchically to allow a separation of the data according to the process modes i ($i = 1, 2, 3, 4, 5, 6$) and groups presented in Section 3.1. Additionally, each group is divided into sub-directories which indicate whether the contained simulations terminated successfully or whether the activated event (fault activation or setpoint change) led to a forced plant-shutdown, leading to the following structure of the data repository:

- Mode i
  - Single fault
    - Simulation completed
    - Simulation stopped
  - Setpoint variation
  - Mode transitions

Each simulation in the repository contains the data described in Section 3.3 and additional metadata associated with the run to enable simple queries of essential simulation parameters, such

Table 5
List of additional measurements introduced by Bathelt et al. (2015).

Legend	Type	Description	Legend	Type	Description
add(1)	meas	Feed A temp. (stream 1) (° C)	add(17)	meas	Conc. of C in stream 2 (mol %)
add(2)	meas	Feed D temp. (stream 2) (° C)	add(18)	meas	Conc. of D in stream 2 (mol %)
add(3)	meas	Feed E temp. (stream 3) (° C)	add(19)	meas	Conc. of E in stream 2 (mol %)
add(4)	meas	Feed A & C temp. (stream 4) (° C)	add(20)	meas	Conc. of F in stream 2 (mol %)
add(5)	meas	Reactor cooling inlet temp. (° C)	add(21)	meas	Conc. of A in stream 3 (mol %)
add(6)	meas	Reactor cooling flow (m^3/h)	add(22)	meas	Conc. of B in stream 3 (mol %)
add(7)	meas	Condenser cooling inlet temp. (° C)	add(23)	meas	Conc. of C in stream 3 (mol %)
add(8)	meas	Condenser cooling flow (m^3/h)	add(24)	meas	Conc. of D in stream 3 (mol %)
add(9)	meas	Conc. of A in stream 1 (mol %)	add(25)	meas	Conc. of E in stream 3 (mol %)
add(10)	meas	Conc. of B in stream 1 (mol %)	add(26)	meas	Conc. of F in stream 3 (mol %)
add(11)	meas	Conc. of C in stream 1 (mol %)	add(27)	meas	Conc. of A in stream 4 (mol %)
add(12)	meas	Conc. of D in stream 1 (mol %)	add(28)	meas	Conc. of B in stream 4 (mol %)
add(13)	meas	Conc. of E in stream 1 (mol %)	add(29)	meas	Conc. of C in stream 4 (mol %)
add(14)	meas	Conc. of F in stream 1 (mol %)	add(30)	meas	Conc. of D in stream 4 (mol %)
add(15)	meas	Conc. of A in stream 2 (mol %)	add(31)	meas	Conc. of E in stream 4 (mol %)
add(16)	meas	Conc. of B in stream 2 (mol %)	add(32)	meas	Conc. of F in stream 4 (mol %)

Table 6
Monitoring of random variation disturbances without influence of measurement noise.

Legend	Type	Flag	Description	Unit
idv_mon(1)	Internal Sim. Value	IDV(8)	Concentration of A in stream 4	mol %
idv_mon(2)	Internal Sim. Value	IDV(8)	Concentration of B in stream 4	mol %
idv_mon(3)	Internal Sim. Value	IDV(8)	Concentration of C in stream 4	mol %
idv_mon(4)	Internal Sim. Value	IDV(9)	Feed component D temp. (stream 2)	° C
idv_mon(5)	Internal Sim. Value	IDV(10)	Feed component A&C temp. (stream 4)	° C
idv_mon(6)	Internal Sim. Value	IDV(11)	Reactor cooling water inlet temp.	° C
idv_mon(7)	Internal Sim. Value	IDV(12)	Condenser cooling water inlet temp.	° C
idv_mon(8)	Internal Sim. Value	IDV(13)	Deviation of reaction kinetics of first reaction	-
idv_mon(9)	Internal Sim. Value	IDV(13)	Deviation of reaction kinetics of second reaction	-
idv_mon(10)	Internal Sim. Value	IDV(16)	Deviation of heat transfer of stipper	-
idv_mon(11)	Internal Sim. Value	IDV(17)	Deviation of heat transfer of reactor	-
idv_mon(12)	Internal Sim. Value	IDV(18)	Deviation of heat transfer of condenser	-
idv_mon(13)	Internal Sim. Value	IDV(20)	Unknown	)
idv_mon(14)	Internal Sim. Value	IDV(21)	Feed component A temperature (stream 1)	° C
idv_mon(15)	Internal Sim. Value	IDV(22)	Feed component E temperature (stream 3)	° C
idv_mon(16)	Internal Sim. Value	IDV(23)	Feed component A pressure (stream 1); flow deviation	kmol/h
idv_mon(17)	Internal Sim. Value	IDV(24)	Feed component D pressure (stream 2); flow deviation	kmol/h
idv_mon(18)	Internal Sim. Value	IDV(25)	Feed component A pressure (stream 3); flow deviation	kmol/h
idv_mon(19)	Internal Sim. Value	IDV(26)	Feed components A & C pressure (stream 4) ;flow deviation	kmol/h
idv_mon(20)	Internal Sim. Value	IDV(27)	Reactor cooling water supply pressure; flow deviation	m^3/h
idv_mon(21)	Internal Sim. Value	IDV(28)	Condenser cooling water supply pressure; flow deviation	m^3/h

as the seed for the random number generator, the initial condition and the magnitude and activation time of process disturbances and set-point changes as well as the average operating cost and production. The seed and initial condition are described by positive integers, where the seed value directly corresponds to the input of the random number generator and the value for the initial condition, which is an integer between 1 and 6 describes the operating mode which was active at the start of the simulation. It directly follows that the initial condition for each simulation run is one of the six operating modes described above. The random disturbances are summarized by a 3x28 matrix, in which the first and second rows include the magnitudes of random disturbances before and after activation, respectively, while the final row specifies the time of disturbance activation in hours. The process setpoints are summarized in similar fashion by a 3x12 matrix, in which the first and second rows contain the setpoint values before and after the change and the final row specifies the time after which the setpoint is changed. For the setpoints, an additional parameter which describes the duration of the setpoint change is included in the metadata. If the duration of the setpoint changes is larger than zero, then the setpoint change follows a ramp profile until it reaches its target value. The provided metadata contains all information needed to re-simulate any run in the dataset.

4. Fault detection

To showcase the use of the provided data, we in this section describe an approach to fault detection in multivariate, cross correlated data as generated by TEP. Various approaches based on chemometrics and machine learning methods have been proposed for this purpose. We focus on a standard approach that relies on dimensionality reduction through Principal Component Analysis (PCA) in combination with two control charts; a Hotelling T^2 chart and a Q-chart. Both charts are evaluated based on their false-alarm rate for in-control process data and the average time until a fault is detected for simulations of all 28 process disturbances.

4.1. PCA-Based statistical process monitoring

PCA is a method that allows for representing a large set of variables by a smaller set of new variables that contains most of the information contained in the original set of variables. PCA reduces dimensionality by iteratively extracting uncorrelated linear combinations of the original variables, a.k.a. principal components (PC). The first PC is the linear combination of all variables that has the maximum variance of all possible linear combinations. The second PC is chosen so that it not only has the second largest variance but is also uncorrelated to the first PC. The procedure continues until from m variables, m uncorrelated PCs are obtained. Since the PCs are selected in decreasing order of variance, if the original variables are highly correlated, a small number of PCs will be able to explain a large portion of the variance in the original data and dimensionality reduction is achieved. The PCA model can be written as

$$X = TP' = \sum_{a=1}^{A} t_a P_a' + E \tag{1}$$

Where X is the matrix with m original variables, T is the so-called scores matrix and P is the loadings matrix of which the columns basically correspond to the weights associated with each variable in the linear combinations. If A (less than m) PCs are retained, the leftover is represented in the residual matrix E. It can be shown that the loadings correspond to the eigenvectors of the covariance matrix of X and the variance of each PC is simply the corresponding eigenvalue of the covariance matrix of X. Data is often standardized prior to the extraction of PCs to account for the scale-dependency of PCA.

When PCA is used in process monitoring, the retained PCs are often summarized under the so-called T^2 statistic and the residuals under Q-statistic as

$$T_i^2 = t_i' \Lambda^{-1} t_i = x_i' P_i \Lambda^{-1} P_i' x_i \tag{2}$$

$$Q_i = e_i' e_i = x_i' \left(I - P_i P_i' \right) x_i \tag{3}$$

where Λ is a diagonal matrix of A eigenvalues in descending order and e_i is the error at the i-th observation vector. Control charts for these statistics are then used to monitor the process performance. This follows a two-phase procedure. In Phase I, data collected from an in-control process is used to estimate the parameters needed to construct the control chart. In this case, data is used to fit the model in equation (1) and subsequently obtain T^2 and Q statistics. Phase II represents the real time monitoring of the process using the model estimated in Phase I and T^2 and Q statistics obtained accordingly. The Phase I control limit for the T^2-chart, based on the beta-distribution, is given by Tracy et al. (1992). The Phase II control limit for the T^2-chart is calculated as

$$\mathrm{UCL}_{T^2} = \frac{A(n+1)(n-1)}{n^2 - nA} F_{\alpha, A, n-A}, \tag{4}$$

where A, n and α are the number of principal components, number of samples and acceptable false alarm rate, respectively. The term $F_{\alpha,A,n-A}$ describes upper α percentile of the F-distribution with A and $n-A$ degrees of freedom. The upper control limit for the Q statistics is given by Jackson and Mudholkar (1979) as

$$\mathrm{UCL}_Q = \theta_1 \left[\frac{z_\alpha \sqrt{2\theta_2 h_0^2}}{\theta_1} + 1 + \frac{\theta_2 h_0 (h_0 - 1)}{\theta_1^2} \right]^{1/h_0}, \tag{5}$$

$$\theta_i = \sum_{j=A+1}^{p} \lambda_j^i, \tag{6}$$

$$h_0 = 1 - \frac{2\theta_1 \theta_3}{3\theta_2^2}. \tag{7}$$

The theoretical control limits listed above assume time-independent data, which has to be taken account in the evaluation of the obtained detection rates. The following sections show that the actual detection rates can deviate from the expected detection rates due to the time dependency in the data.

4.2. Process surveillance using TEP simulation data

In this section, we show how the provided data can be used to evaluate process performance based on PCA and using T^2 and Q charts. For this, we propose to use the Average Run Length (ARL) as the metric. A run length is defined as the number of observations it takes to observe an out-of-control signal for a given control chart. Hence ARL represents the expected number of observations before the declaration of a signal. The in-control ARL (ARL_0) is desired to be as large as possible to avoid false alarms whereas the out-of-control ARL (ARL_1) should be as small as possible to avoid delay in detection. Unfortunately, it is not possible to achieve both of these goals simultaneously as they counteract each other. Therefore, the common practice is to set the tolerable false alarm rate (or $1/\mathrm{ARL}_0$) at a nominal value such as 5% and observe the performance of the proposed control chart for detecting a shift. It should be noted that the estimation of ARL requires multiple simulation runs of the same scenario and hence cannot be achieved if only one simulation run is available as provided by Braatz (2020). The researchers are then forced to use alternative metrics such as Fault Detection Rate, which is a flawed metric for various reasons such as it only represents a single realization of the process behavior and depends on the size of the operating window for which the control chart is used. Therefore, we strongly recommend the researchers to employ datasets that provide replicated simulations runs for the same scenario and compare any methods based on appropriate metrics.

4.3. Phase I analysis

Before we delve into the fault detection, we will first investigate whether the control charts we employ operate at nominal false alarm rates, considering the continuous process measurements and manipulated variables (pd(2–23) and pd(43–54)), excluding any signals that remain constant during simulation runs. For that, we use the first 500 observations of the simulations of all faults at maximal fault magnitude for each mode. This generates 5600 simulation runs at 500 observations each for an in-control process, given 200 repeat simulations for each of the 28 process faults. Since the respective process faults are always introduced after 600 observations, the first 500 observations always represent nominal, fault-free operating conditions, thus making them suitable for Phase I analysis. We then record the first time each of these datasets declares an alarm as the run length for that dataset. Then ARL_0 is obtained as the average of these run lengths. We do this for both T^2 and Q charts using a nominal false alarm rate (α in (6) and (7)) of 5%, which corresponds to an ARL_0 of 20. For cumulative variance explained in PCA, we use 75%, 80%, 85% and 90%. It should be noted that in many of previous studies using TEP simulation data, PCA often reveals that 14 PCs are sufficient to explain 85% of the total variance. In Table 7, we provide the cumulative percentage of variance (CPV) and corresponding average and standard deviation of the number of PCs for all modes. It can be seen that for 85% CPV, around 18PCs are needed for most of modes. Moreover 14 PCs correspond to 75% CPV. This slight discrepancy can be attributed to the control scheme as well as the number of variables considered in the analysis being different from previous studies. We therefore recommend the researchers to take this into consideration when using the provided datasets. In Table 8, we report the ARL_0 for both charts for all modes for all CPV options. As noted earlier, ARL_0 for both T^2 and Q charts is expected to be 20 for the false alarm rate of 5%. It can be seen that while the ARL_0 for the Q chart is around 20 for all modes, the ARL_0 for T^2 chart is not always at the nominal. For instance for Mode 1, which is the most commonly used mode, ARL_0 is higher than expected yet varies with different CPV options. One explanation can be that UCL

Table 7
Number of principal components for Phase I PCA models.

	Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Mode 6	
Var expl.	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ	μ	σ
75%	13.91	0.37	13.91	0.39	14.39	0.50	12.53	0.65	13.04	0.77	13.97	0.93
80%	15.69	0.46	15.71	0.48	16.07	0.35	14.21	0.66	14.74	0.78	15.72	0.95
85%	17.59	0.50	17.61	0.52	18.01	0.29	16.02	0.67	16.58	0.78	17.63	0.95
90%	19.70	0.46	19.73	0.48	20.10	0.37	18.07	0.68	18.63	0.78	19.82	0.95

Table 8
ARL_0 for the in-control PCA model for based on healthy simulation data.

	Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Mode 6	
Var expl.	T^2	Q	T^2	Q	T^2	Q	T^2	Q	T^2	Q	T^2	Q
75%	29.64	21.42	21.40	20.20	36.10	23.75	33.67	18.73	34.37	21.62	15.03	19.71
80%	28.54	21.36	21.39	20.49	34.66	24.31	32.29	18.53	33.82	21.33	14.59	17.72
85%	27.83	21.64	21.13	20.34	34.36	24.32	31.21	18.54	33.15	22.20	13.91	17.36
90%	27.46	22.13	21.05	20.05	33.56	25.12	31.02	19.16	32.76	22.09	13.46	18.31

Table 9
ARL1 for the 28 process faults for T^2 and Q charts.

	Mode 1		Mode 2		Mode 3		Mode 4		Mode 5		Mode 6	
	T^2	Q	T^2	Q	T^2	Q	T^2	Q	T^2	Q	T^2	Q
IDV1	2.8	2.2	2.9	2.1	2.8	2.3	1.5	1.0	3.2	2.2	2.9	2.1
IDV2	6.2	3.1	5.5	3.0	8.5	4.8	2.1	1.1	4.8	3.2	7.3	4.6
IDV3	26.6	4.4	34.0	7.6	14.7	3.0	17.6	1.7	23.5	7.9	11.8	3.5
IDV4	1.9	1.8	1.9	1.9	1.9	1.8	1.0	1.0	1.9	1.8	1.8	1.6
IDV5	5.6	3.0	3.8	3.1	1.9	1.8	1.0	1.0	3.5	2.4	1.8	1.8
IDV6	1.9	1.8	1.9	1.8	1.9	1.8	1.0	1.0	1.9	1.8	1.8	1.7
IDV7	1.9	1.8	1.9	1.8	1.9	1.8	1.0	1.0	1.9	1.8	1.7	1.7
IDV8	11.9	6.2	11.2	6.1	10.6	5.4	8.8	5.2	10.5	6.4	9.5	4.8
IDV9	23.0	6.8	31.5	7.6	15.5	6.4	17.4	6.8	26.9	6.6	15.0	6.3
IDV10	17.4	6.1	17.4	6.5	13.9	6.1	20.7	6.9	16.7	6.8	14.0	5.7
IDV11	5.3	3.9	5.4	4.0	5.1	4.0	3.3	2.6	5.4	3.9	4.7	3.4
IDV12	9.74	5.3	9.3	5.2	4.4	3.2	4.8	3.4	8.5	5.7	4.6	3.1
IDV13	18.0	6.3	16.5	6.4	19.1	6.9	16.3	6.2	13.6	5.9	15.1	7.2
IDV14	2.7	2.2	2.3	2.0	2.6	2.4	1.3	1.3	2.2	2.0	2.8	2.2
IDV15	26.5	6.8	25.6	7.3	26.0	7.1	26.8	6.9	24.2	6.8	26.6	6.5
IDV16	32.1	7.6	32.0	7.7	23.8	6.8	25.9	7.7	26.4	7.0	24.9	7.9
IDV17	15.7	6.4	15.4	6.9	15.0	6.1	15.3	6.2	14.6	6.9	13.5	5.9
IDV18	23.4	6.5	23.1	7.6	20.1	6.9	23.3	7.8	21.5	7.0	17.0	6.9
IDV19	6.0	3.6	6.3	3.4	4.2	3.0	9.6	3.2	9.6	3.8	5.5	3.1
IDV20	21.4	7.2	18.8	5.7	18.4	5.9	17.7	7.1	16.9	7.5	14.9	5.8
IDV21	25.8	7.5	30.6	7.0	28.8	6.3	37.7	7.5	25.2	8.1	28.3	6.8
IDV22	20.3	6.8	16.6	6.6	27.5	7.0	15.2	5.7	14.5	6.2	30.8	7.1
IDV23	28.8	6.3	27.7	6.3	29.1	6.8	25.6	5.6	23.8	5.6	21.2	5.6
IDV24	10.5	5.6	21.1	6.1	9.2	5.0	8.0	4.8	14.7	6.7	7.1	4.6
IDV25	13.1	5.3	10.3	5.4	22.8	5.9	9.7	4.9	8.5	6.1	18.4	5.4
IDV26	14.9	5.5	16.4	6.8	17.1	6.6	13.8	5.9	15.2	6.6	12.2	5.1
IDV27	4.9	3.6	5.6	3.9	4.9	3.8	3.0	2.5	5.4	4.1	4.6	3.4
IDV28	25.7	6.8	21.3	6.2	10.2	3.9	11.0	5.3	20.5	6.72	16.9	4.9

in Equation (4) is based on the assumption that the observations are uncorrelated in time. This is usually violated when data is collected in time as in continuous processes let alone in TEP simulations. But this is beyond the scope of this paper. Further details can however be found in Bisgaard and Kulahci (2005); Vanhatalo and Kulahci (2015a,b); Vanhatalo et al. (2017). For a more practical solution, the provided data offer once again a viable solution. Instead of using Equation (4) to calculate the UCL for the chart, by varying the UCL systematically, the right UCL corresponding to the desired false alarm rate can be found and used in both Phase I and II.

4.4. Phase II analysis

In this section, we investigate the fault detection performance of and Q charts. In Table 9, we provide the ARL1 for all faults and all modes for 85% CPV in PCA. It should be noted that for all faults and modes, the ARL1 for the Q chart is reasonably low. Yet for the T^2 chart, we observe a great variation in ARL1. This is particularly of concern when ARL1 is similar to (or even higher than) the nominal false alarm rate corresponding to ARL_0 from Table 8. This in fact simply implies that the control chart is not able to catch the faults properly. It should be noted that nature of the faults and their impact to the system vary substantially. If a faults impact is isolated to a part of the process or taken care of by a controller for example, multivariate schemes as proposed in this article may have a hard time detecting these faults as these schemes are using aggregate measures such as principal components and Hotelling T^2 statistics over all variables, most of which will not be affected by such an isolated fault. For example, in many previous studies some faults such as IDV 3 and IDV15 are deemed notoriously hard to catch with simple schemes as we employed in this paper (Liu et al., 2015). Of the faults of type *step* (IDV1-IDV7), IDV3 shows significantly worse fault detection performance than the other faults, while multiple of the faults of type *random variation* and *sticking* are hard to detect using T^2 statistics (IDVs 9, 13, 15, 18, 20–23). We expect the provided datasets to offer ample opportunities for researchers to test newly develop methods towards improving the fault detection rates, particularly for these faults.

5. Conclusion

We present a new reference dataset for the Tennessee Eastman process that is based on the most recent, and most versatile, implementation of the Tennessee Eastman simulator by Bathelt et al. (2015). Acknowledging the necessity of large amounts of data for the development of statistical process monitoring and machine learning based fault-detection methods, the new dataset provides a substantially larger amount of data points for each operating mode of TEP than any currently openly available dataset. Furthermore, all six modes of operation that were originally presented by Downs and Vogel (1993) are represented, while current datasets focus exclusively on the analysis of the first mode of operation. For fault analysis, all 28 available process faults are simulated 500 times with independent random seeds, providing data for eight additional faults that are not represented in most currently available datasets. The feature of the simulator by Bathelt et al. (2015) to scale the magnitudes of faults is exploited to provide runs with lower fault magnitudes for all fault scenarios. To establish a baseline for the expected performance of statistical process monitoring approaches, we provide a benchmark for detection performance by applying a PCA-based statistical monitoring approach together with Hotelling T^2 and Q-charts, for which we report detection performance using average run length as the performance metric. To provide a larger amount of data that represents nominal operation, process configurations in regions around the original six operating modes as well as transitions between the operating modes are simulated.

Given the extent and variation of the new dataset, we believe that it provides a more viable benchmark than any other currently available dataset and thereby enables continued, improved method development that leans on the Tennessee Eastman process as a case study.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Christopher Reinartz: Conceptualization, Software, Validation, Investigation, Writing - original draft, Writing - review & editing. **Murat Kulahci:** Conceptualization, Validation, Formal analysis, Writing - original draft, Writing - review & editing. **Ole Ravn:** Resources, Supervision, Project administration.

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